

THANH LONG LY

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EDUCATION

National Economics University (NEU)

Bachelor of Data Science in Economics and Business (CGPA: 3.55/4.0)

Hanoi, Vietnam

Aug 2023 – Aug 2027

EXPERIENCE

Alpha Data Academy

Agentic AI Engineer

Hanoi, Vietnam

Jun 2025 – Present

- Achieved a **98% performance score** for Rang Dong by architecting a **LangGraph** multi-agent system (Ralli), implementing a **hybrid retrieval strategy** (semantic + keyword) to bridge technical jargon gaps.
- Integrated **LangGraph Persistence** with **Supabase PostgreSQL**, developing "think-before-talk" logic that optimized multi-turn inference latency.
- Supported **100+ concurrent requests** by architecting an asynchronous **FastAPI** backend, orchestrating containerized services with **Docker**, and implementing observability via **Grafana** and **Opik**.

Predictive Models and ML Research Application Lab (NEU)

AI Research Assistant

Hanoi, Vietnam

Jan 2025 – Present

- Synthesized state-of-the-art Agentic AI frameworks by analyzing **50+** academic papers to guide architectural decisions.
- Co-authored a comprehensive research report on Agentic AI adoption, translating technical metrics into business impact for stakeholders.

FEATURED PROJECTS

ONMAI.AI - Real-Time Multimodal AI for IELTS Assessment

Independent Full-Stack Developer

Jan 2026 – Present

- Developed an AI platform for IELTS learners, providing feedback grounded in **Band Descriptors** through multimodal analysis.
- Architected a real-time voice pipeline using the **Gemini Live API via WebSockets**, ensuring sub-100ms latency for a natural experience.
- Implemented conversational memory to maintain strict contextual consistency across multi-turn language evaluations.

SRGNN - Session-Based Recommender System (MLOps)

Infrastructure Lead

Mar 2026 – May 2026

- Implemented a **Graph Neural Network (GNN)** for real-time recommendations, optimizing for anonymous users in a cookieless environment based on current session behavior.
- Orchestrated an end-to-end MLOps pipeline using **GitHub Actions** for CI/CD, pushing containerized images to GHCR and deploying to **Amazon EKS**.
- Managed cloud infrastructure with **Terraform (IaC)**, integrating **DynamoDB** and **S3** for data storage, ensuring high availability and horizontal scaling.

RIA - Agentic Robo-Investment Advisor (Swin Hackathon 2026)

Tech Lead

Feb 2026 – Mar 2026

- Designed a **hierarchical MAS** with a session router directing requests to three stateful agents, ensuring functional transparency.
- Mitigated hallucinations by isolating quantitative calculations into **deterministic subagents**, utilizing the LLM for orchestration.
- Proposed **AWS-native infrastructure** (Bedrock, RDS, Redis) for financial compliance and sub-millisecond state management.

Gigabyte Sale Agent (Master AI Hackathon 2025)

Lead Architect

Oct 2025 – Dec 2025

- Won **1st Place** by building a high-precision agent for Gigabyte, reasoning through inquiries and complex **ROI calculations**.
- Engineered a **hybrid agentic workflow** with a central router for state transitions, improving groundedness and flexibility.
- De-coupled ROI logic from the LLM by developing a **deterministic calculation tool**, enhancing accuracy in live demos.

SKILLS & INTERESTS

- Technical Stack:** Python, SQL, Docker, AWS, Git, FastAPI.
- AI/ML Stack:** PyTorch, TensorFlow, HuggingFace, LangChain, LangGraph, Scikit-learn, Qdrant, Opik, Grafana.
- Core Expertise:** AI Engineering (Agentic Workflows), MLOps (Deployment Loops), Full-Stack Development (Architectural Modularity), Product Strategy (Analytical Problem Solving).
- Interests:** Agentic AI Research, Applied Applications, Competitive Hackathons.